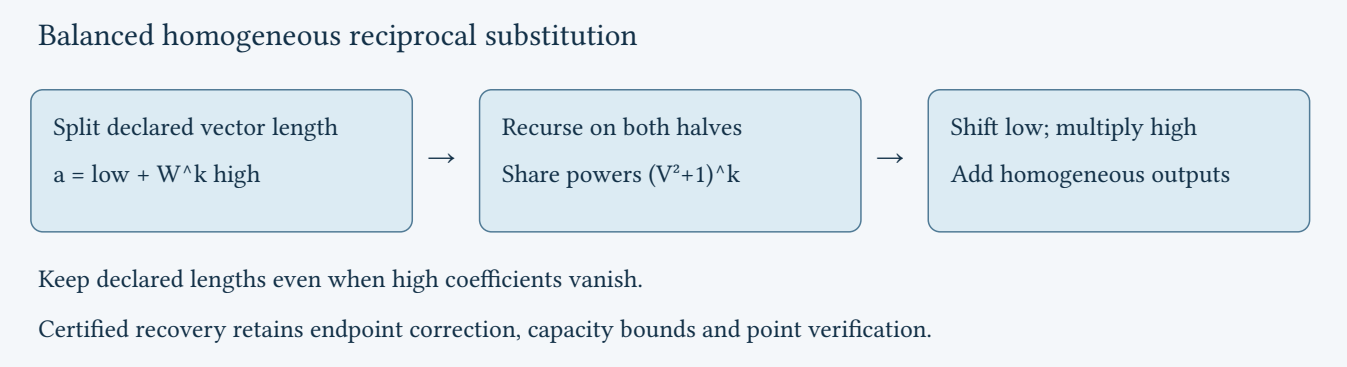
Balanced reciprocal substitution

T-1923 · Prime fields · 2026-11-08

The optional recovery adapter replaces the reciprocal lift's growing coefficient loop with balanced homogeneous recursion. Character-domain provenance, phase-free initialization, endpoint multiplicity correction and final point checks are retained.



Recurrence. For a declared degree d and split $a=low+W^k$ high, define $R_a=V^d a(V+V^{-1})$. Then $R_a=V^{(d-k+1)}R_{low}+(V^2+1)^k R_{high}$. Split coefficient lengths in half, reuse powers within one transform, and retain declared lengths even when high coefficients vanish.

Arithmetic scope. With regular multiplication cost M(n), the standard balanced bound is O(M(n) log n). The former dense coefficient loop was quadratic. No timing crossover is selected. Cached powers are copied; shifts, map operations, polynomial scratch storage, allocations and output remain charged. Initial character domains still grow with L.

Qualification. 1,728 coefficient equalities and 6,912 point identities pass over four fields, with input lengths through 1,025 and zero/sparse/dense patterns. 641 phase-certified lift comparisons, 1,282 endpoint controls and 641 cap controls pass. All 3,600 public cases recover correctly, including 228 odd-L cases; 3,372 predecessor cases retain outcomes, routes and source statistics. PCC-20261108-1228b4 is unsigned producer evidence.

Custody. The fresh build and focused/inherited checks pass under 90/60/60-second bounds. No failed attempt occurred. 140 copied sources and 157 compiler dependencies are hash-bound; inherited output is byte-identical. The large-prime polynomial fixture is not a large-prime recovery run. Independent signed review pending.

Next construction. An observation-derived low-degree polynomial could allow phase intersection using D_L modulo that polynomial, avoiding dense D_L construction. Producing such a useful polynomial from public observations without degree-L work remains the missing construction; fixture-derived polynomials do not establish it.

Sources. Exact predecessor certificate/lift modules and retained NIST DLMF18.5.1–2 reciprocal identities. No novelty or timing claim.